

TASK 1 – REACT BASICS

Read and prepare answers for:

1. What is React?

React is a single page application.

React is a javascript library used to build a interactive web applications

React has a component based architecture

React was introduced by meta ,at 2004.

2. Why do we use React?

React is helps to reuse the code because of the component based architecture

Reusability is very high and faster to build

3. Is React a library or framework?

React is a library that contains pre defined classes .

Can be installed anytime and use the predefined classes .

4. What are the main features of React?

- Component based Architecture
- One-way dataflow
- Single -page -application
- Virtual Dom
- Dynamic Rendering
- Reconfilation
- Diffing Algorithm

5. What is a Single Page Application (SPA)?

React is a single page application helps to build or repair a component without reloading the entire web page .One of the main benefits of react.

6. What is declarative UI?

Describe what the ui should be

Easier to read

Framework updates the DOM

TASK 2 – COMPONENTS

Read and prepare answers for:

1. What is a Component in React?

Each Component is a file contains jsx code in it. It is a small piece of reusable code .

2. Why do we use Components?

Components is a collection of multiple reusable piece of code.

It helps to manage a manageable , small, reusable piece of code

3. What is a Functional Component?

Functional component is a specific type of component that written as a javascript function and returns jsx.

functional component supports hooks which is simple to write code.

4. What is Component-Based Architecture?

Component-Based Architecture is a design approach where an application is built using small, reusable, and independent components. Each component has its own responsibility, making the application easier to develop, maintain, and reuse.

5. What is a Component Tree?

A Component Tree is the hierarchical structure of React components, showing the parent-child relationship between components in an application.

6. Why should component names start with a capital letter?

React uses capitalization to distinguish components from HTML elements.

For example: <Header/> ,<header>

7. What is a reusable component?

A reusable component is a component that can be used multiple times in different parts of an application without rewriting the same code.

TASK 3 – JSX

Read and prepare answers for:

1. What is JSX?

Javascript xml is a language used to write code inside a react component

2. Why do we use JSX in React?

To write the html inside the javascript is called as JSX
It makes easy to read and understand the code

3. Is JSX HTML?

No,jsx is different from html .
jsx contains html and javascript.
It is a JavaScript syntax extension that looks like HTML. React converts JSX into JavaScript.

4. How do we write JavaScript expressions inside JSX?

We write JavaScript expressions inside curly braces {}.
Function fruits(){
Return fruits;
}

5. Why do we use className instead of class?

We use className because class is a reserved keyword in JavaScript. In JSX, className is used to apply CSS classes to HTML elements.

6. What is React Fragment?

A React Fragment is a special component that lets us group multiple elements without adding an extra HTML element to the DOM.

7. What are the important JSX rules?

JSX must return one parent element.
Close all tags.
Use className instead of class.
Write JavaScript expressions inside {}.
Component names must start with a capital letter.
Use camelCase for HTML attributes.

TASK 4 – DOM, VIRTUAL DOM & RECONCILIATION

Read and prepare answers for:

1. What is DOM?

DOM (Document Object Model) is a tree-like representation of an HTML document. It allows JavaScript to access and modify HTML elements dynamically.

2. What is Virtual DOM?

The Virtual DOM is a lightweight copy of the Real DOM maintained by React. React updates the Virtual DOM first, compares it with the previous version, and then updates only the changed parts of the Real DOM.

3. What is the difference between DOM and Virtual DOM?

DOM (Real DOM)	Virtual DOM
Actual HTML page in the browser	Lightweight copy of the Real DOM
Directly updated	Updated first by React
Slower for frequent updates	Faster because React updates only the changed parts
Managed by the browser	Managed by React

4. What is Reconciliation in React?

Reconciliation is the process React uses to compare the old Virtual DOM with the new Virtual DOM and determine what has changed. It then updates only those changes in the Real DOM.

5. What is Diffing?

Diffing is the comparison process used during reconciliation. React compares the old Virtual DOM with the new Virtual DOM to find the differences.

6. How does React identify changes in the UI?

React identifies UI changes by comparing the old and new Virtual DOM using the diffing algorithm, then updates only the changed parts of the Real DOM.

7. Why is key important while rendering lists?

A key is a unique identifier given to list items. It helps React identify which items are added, removed, or updated, making rendering faster and more efficient.

TASK 5 – REACT RENDERING PROCESS

Read and understand this complete flow:

Prepare answers for:

1. What is Rendering?
Rendering is the process of converting a React component (JSX) into React elements and displaying the UI in the browser.
2. What is Re-rendering?
Re-rendering is the process where React renders a component again after its state or props change.
3. What causes a component to re-render?
A component re-renders when its state changes, its props change, its parent re-renders, or the context it uses changes.
4. Does re-render mean the complete browser DOM is recreated?

No. During re-rendering, React updates only the necessary parts of the Real DOM instead of recreating the entire DOM.

5. What happens during reconciliation?
During reconciliation, React compares the old and new Virtual DOM, finds the differences, and updates only the required parts of the Real DOM.
6. What is the relationship between Virtual DOM, Diffing and Reconciliation?
7. The Virtual DOM stores the UI, Diffing compares the old and new Virtual DOM, and Reconciliation updates only the changed parts of the Real DOM.
8. Explain the React rendering process in your own words.

React starts by rendering a component and creating the Virtual DOM. When the state or props change, React creates a new Virtual DOM, compares it with the previous one using Diffing, performs Reconciliation, updates only the changed parts of the Real DOM, and finally displays the updated UI in the browser.